Phase 1 Report

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Abstract

Our goal for the project is to provide a simple management system for a new, small restaurant business. Our feature targets are expense and revenue tracking, as well as inventory analytics and statistics. We want this software to be intuitive and easy to use. If given time, we can implement requirements for other industries via plugins to extend our user-scope. The biggest challenge we will run into is time-management, but another potential issue is gathering all the requirements of our target user. We plan on using HTML, CSS, SQL, and Python for implementing this software, adding Bootstrap and JavaScript to increase functionality if time permits. To best match our project's goal, we are using an open-source license to keep availability and transparency high.

Project Description

The main goal of the project is to provide a simple management system for a small restaurant business. An issue with startup restaurants is their need for software to assist in management, but most lack the funding to purchase and use a commercial on-the-shelf software solution. Another issue with most commercial on-the-shelf software solutions for restaurant management is their complexity. Often, these software solutions are too advanced, which causes difficulty in implementation as well as usability. Our goal with our software is to make it easier for the owner of the restaurant to stay organized and keep track of important business information.

Main Functionality and Characteristics

The goal of this software is to provide the user with choice on which aspect of restaurant management that they want handled by software and which aspects that they want to manage traditionally. Our goal is to provide features like expense and revenue tracking for budgeting purposes. We also want to include inventory analytics and statistics.

Intended Users and Key Usability Goals

Our intended users for this software are first-time business owners working in the food industry. We need to create software that is easy to use and configure for those without a deep understanding of the functionality of computers. We also know that business owners are busy, therefore, this software needs to be responsive and able to run without much user input.

Further Development Beyond Class

There are a lot of different avenues that this software can take. We can start looking into requirements needed for first-time business owners in other industries. This could include client management, better financial management, and employee schedule management. These kinds of features are not required for the food industry but having them implemented in our software would open the doors to more users across various industries.

To avoid running into problems with the software being too complex for the average user, we can implement these features through plugins. Then, a business owner would only need the slim base package and the features their industry needs through the available plugins. This would add large amount of flexibility to the program while maintaining its simplicity.

Potential Challenges

Given that this is one of the first large projects that our group members have worked on, there are a lot of potential challenges that could arise. The most daunting challenge would be time management. There is a lot of new stuff that our group members will be working on for the first time, and it will take us time to learn new languages and then implement them. Another challenge will be working with the end user on narrowing down the exact requirements they need out of the software. A first-time business owner is going to be under a lot of stress; they may not know everything they are struggling with (and therefore need the software to help automate) until later in the project.

Intended Platforms, Languages, Libraries, and Tools

We intend to implement this software as a web-site application. This grants flexibility and the ease-of-entry that the end user is requesting. We plan on doing so by using HTML and CSS to create the webpages as well as have finer control over the layout and appearance of the site. We will then use SQL and Python for the back-end of the software. SQL will handle interfacing with our database system, which will be especially useful for our inventory management portion, while Python will help with calculations and bridge the gap between the web interface and our database.

If time permits, we can also implement other languages to add functionality to our product. Bootstrap can help patch up and clean the interface on the site where HTML and CSS lack. We can also implement JavaScript functionality to help the interactivity of the website.

Project Helping Professional Growth

A project like this helps all of its members for the future. It provides a great first step into applying what we have spent years learning in a classroom environment to a real-world project. The mistakes and lessons learned from them will help provide a level of learning that is not available in a classroom setting. Also, a project like this looks great on a resume, as it shows employers that you have experience not found in a typical work or education environment.

Team Member Backgrounds

Our three team members have various backgrounds in the languages and environments that they are familiar with, therefore, we are assigning roles within the project that best match our team members' strengths.

Julius is strong with Python as well as SQL, therefore, he will be working on the financial side of the back-end, also be working on report generation. Jorden has some experience with Python, but is more comfortable with HTML and CSS, and will be working on the front-end as well as the user

interface design. Cole has a lot of experience with Python, as well as hints of experience with SQL, and has been assigned the back-end and the inventory solution of the system.

Open-Source Significance

We plan on licensing our software as open-source. That allows the software to be used by all and freely modified by end users who may need minor tweaks for it to better fit into their needs. We believe an open-source approach aligns better with first-time business owners who want transparency and freedom within their software solutions.

Solutions like this already exist for businesses with larger budgets and loftier goals, but for our target user group, the selection narrows. One close example is Haris Muneer's Restaurant Management System, which was also developed as a Software Engineering project. While the idea is similar, the scope is much larger, and the way the software is implemented is different. His software covers restaurant management at a higher level, working on ordering, billing, kitchen, hall, and inventory management. Also, his software is implemented as an Android app, which limits the availability of the software (Muneer, 2018/2026).

Project Related Resources

Bartoszek. (2026, May 18). *Top 10 All-In-One Restaurant Management Software in 2026 (Tried & Tested)*. UpMenu. https://www.upmenu.com/blog/restaurant-management-software/

Jenkins, A. (2025, January 16). *How to Best Manage Your Restaurant Inventory*. Oracle NetSuite. https://www.netsuite.com/portal/resource/articles/inventory-management/restaurant-inventory-management.shtml

Muneer, H. (2026). *Harismuneer/Restaurant-Management-System* [Java]. https://github.com/harismuneer/Restaurant-Management-System (Original work published 2018)